



Abstract

We give an elementary calculation of the Euler characteristic of the boundary complex of the moduli space of genus-one stable curves. Our result agrees with the work of Chan–Galatius–Payne, who proved that this boundary complex is homotopy equivalent to a particular wedge of spheres, but the proof technique in our case is purely combinatorial: we construct a sign-reversing involution on genus-one boundary strata modeled on a similar calculation in genus-zero due to Blankers–Gillespie–Levinson.

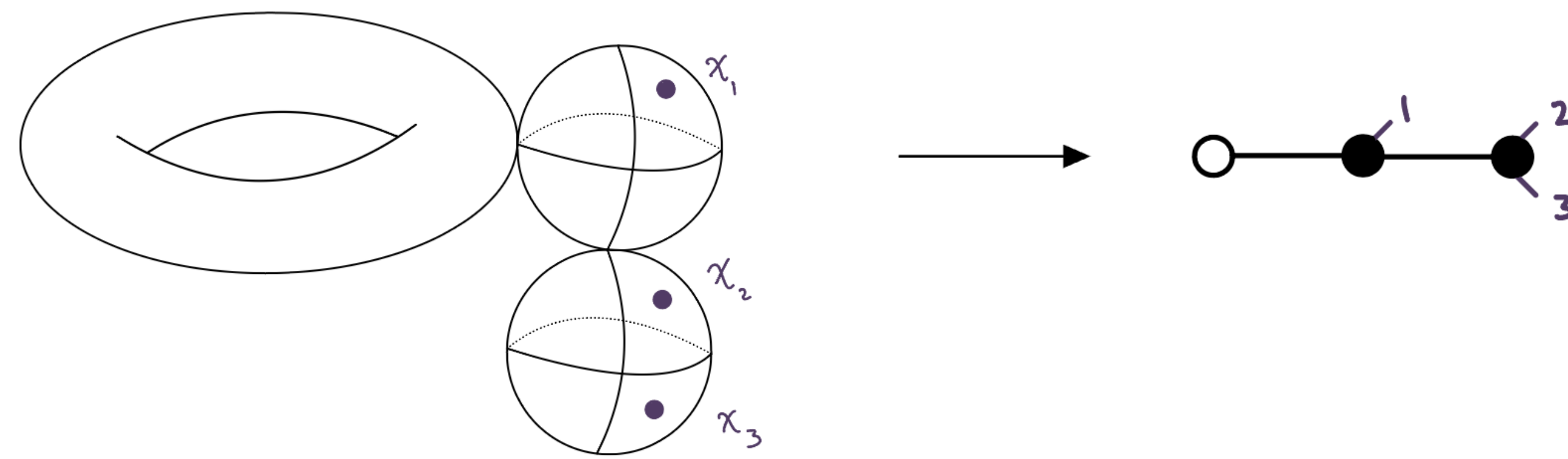


Figure 1. A genus-one curve and its dual graph. Genus-zero vertices are filled circles, genus-one vertices are empty circles.

Moduli Space of Curves

We denote the *Deligne–Mumford* moduli space of stable n -pointed, genus- g curves by $\overline{\mathcal{M}}_{g,n}$. Points in $\overline{\mathcal{M}}_{g,n}$ are in bijection with tuples $(C; x_1, \dots, x_n)$, where C is an algebraic curve over $\mathbb{C}$ with genus g and at worst **nodal singularities**, and $x_1, \dots, x_n \in C$ are distinct **marked points**.

Associated to each tuple is a **dual graph** (as seen in Figure 1) with the following data

1. A **vertex** v for each irreducible component of C , labeled according to its genus.
2. An **edge** between vertices for each node between corresponding irreducible components.
3. A **half-edge** for each marked point labeled by the index in $\{1, 2, \dots, n\}$.

Of interest to us is the quantity,

$$\chi_{g,n} := \sum_{\substack{\text{dual graphs } G \\ \text{for } \overline{\mathcal{M}}_{g,n}}} (-1)^{\#E(G)}.$$

Main Theorem

For $n \geq 3$

$$\chi_{1,n} = (-1)^n \frac{(n-1)!}{2}.$$

Sign-Reversing Involutions

A **sign function**, which we denote σ , is a function that maps elements from a set X to ± 1 . An **involution** is a function f on a set X such that for any $x \in X$ we have $f(f(x)) = x$; In other words, applying f twice gives the identity function.

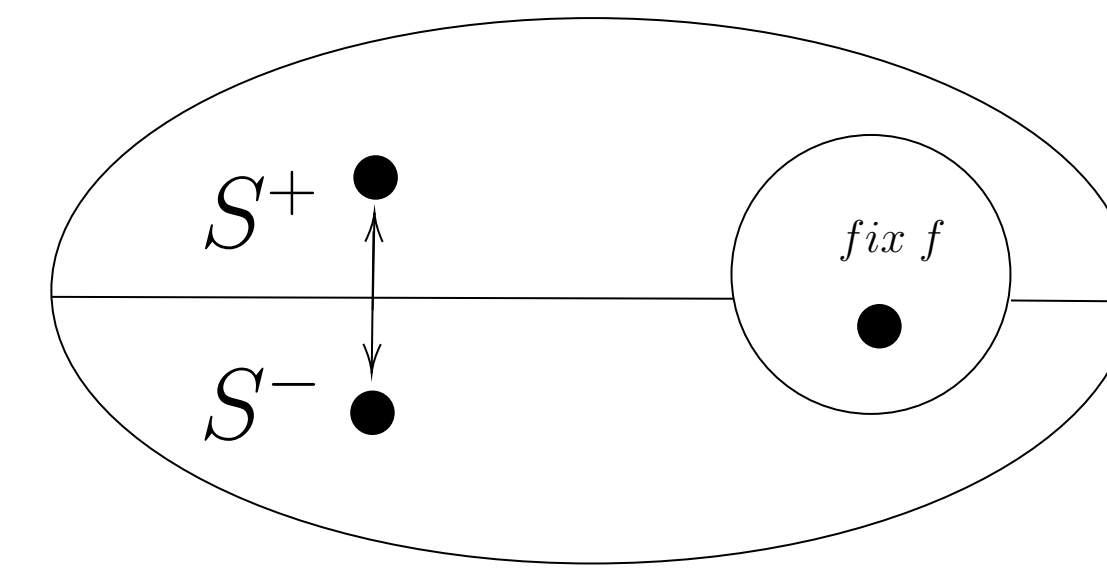


Figure 2. Sign reversing involution on a set S

As Figure 2 suggests, if $\sigma(x) = +1$, (the sign of x is 1), then $\sigma(f(x)) = -1$. Simply put, applying f to an element swaps its sign, unless the element belongs to the **fixed set**. This is just the set of points of X such that $f(x) = x$. As one may see, given an alternating sum that enumerates some set S , (such as the sum in our main Theorem), we can devise a sign-reversing involution on S , allowing us to count the much smaller fixed set.

Why Do We Care?

- For a dual graph G , the **boundary stratum** S_G denotes elements of $\overline{\mathcal{M}}_{g,n}$ with dual graph G or whose dual is a **degeneration** of G .
- The **boundary complex** is constructed by taking a d -simplex for every stratum with $\text{codim}_{\mathbb{C}}(S_G) = d + 1$ and gluing according to edge contractions, or equivalently, set containment $S_G \subseteq S_H$.
- The work of [3] shows this complex is homotopy equivalent to a wedge of

$$\frac{(n-1)!}{2}$$

spheres of dimension $n - 1$.

- We have

$$\chi(\Delta_{g,n}) = \sum_{i=0}^{3g-4+n} (-1)^i \# \{(i+1)\text{edge dual-graphs for } \overline{\mathcal{M}}_{g,n}\},$$

which implies

$$1 - \chi(\Delta_{g,n}) = \chi_{g,n}.$$

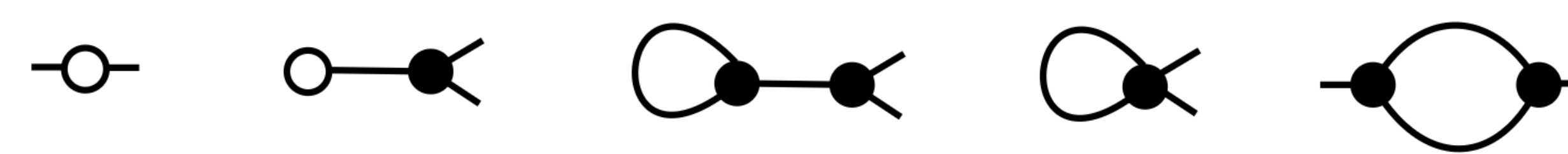


Figure 3. The dual graphs for $\mathcal{M}_{1,2}$

Proof Sketch

Fix $n \geq 3$ and let $S_{1,n}$ denote the set of dual graphs for $\overline{\mathcal{M}}_{1,n}$. If G contains a unique cycle C as a subgraph then for each vertex v we write $L_v \subseteq [n]$ to be the set of labels of vertices which map to v under the contraction of edges in $E(G) \setminus E(C)$. We define our involution as follows:

1. If G has a genus-one vertex v with $\text{val}(v) = 1$, then contract G along its unique incident edge e .
2. If G has a genus-one vertex $\text{val}(v) > 1$, we change the genus of v to zero, and form an edge between v and a new genus-one component with no marked points.
3. If G has a cycle C as a subgraph there exists v such that $\#L_v = 1$ for all v we fix G .
4. If G has a cycle C as a subgraph there exists v such that $\#L_v > 1$, then among such v choose the one with smallest minimum element. Viewing v as vertex in $E(G) \setminus E(C)$ if $\text{val}(v) = 1$, we contract along its unique incident edge e .
5. In G is as before, then we replace v with vertices v_1, v_2 joined by an edge e , where v_1 is incident to the two unique edges in C and e , and v_2 inherits the remaining information of v .

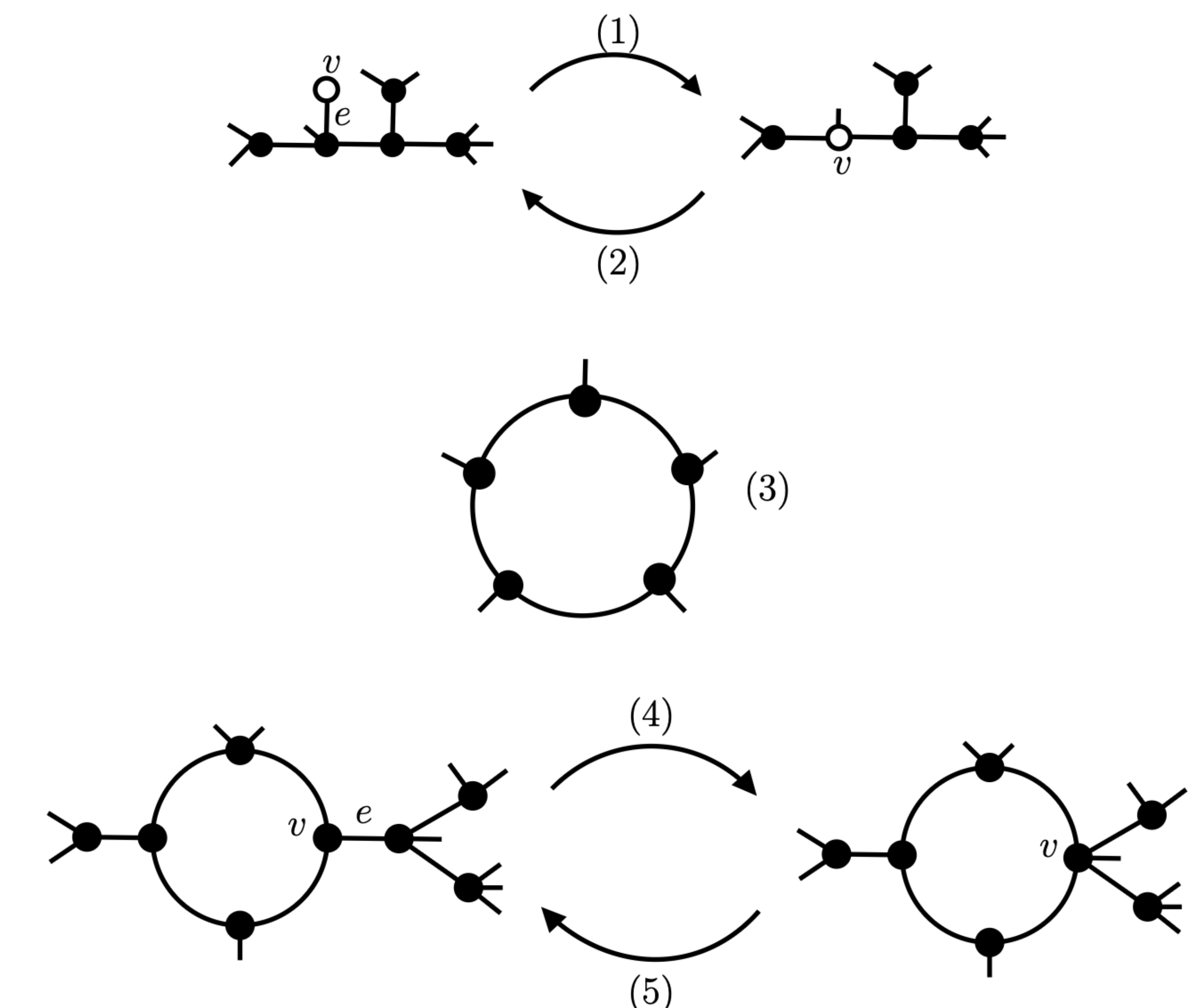


Figure 4. An illustration of cases (1)–(5) with labels on half-edges omitted

References

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